CSC3041F

Control

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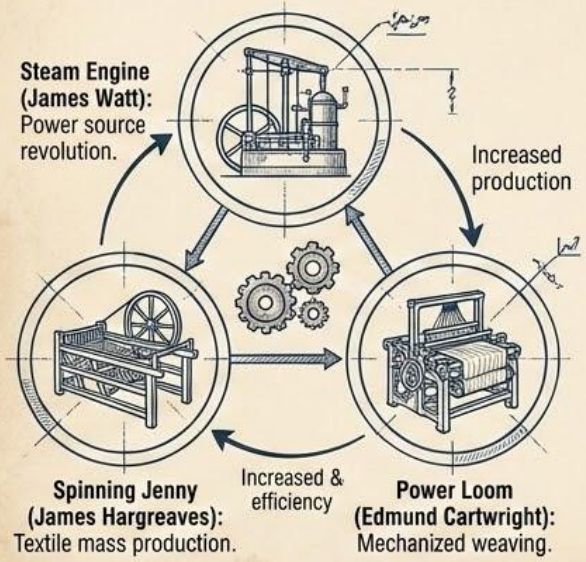
Image: Lucie de Moyencourt

THE INDUSTRIAL REVOLUTION: A GLOBAL TRANSFORMATION

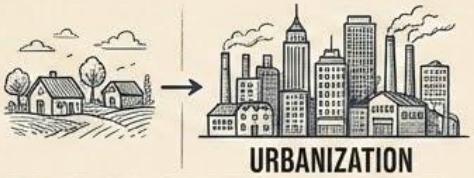
From manual labor to machine-driven production. Key milestones and innovations.

1760 — 1840

KEY INNOVATIONS & IMPACT



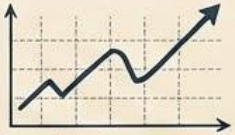
SOCIOECONOMIC SHIFTS



FACTORY SYSTEM



STANDARD OF LIVING



Long-term improvement, short-term hardships.

GLOBAL DIFFUSION



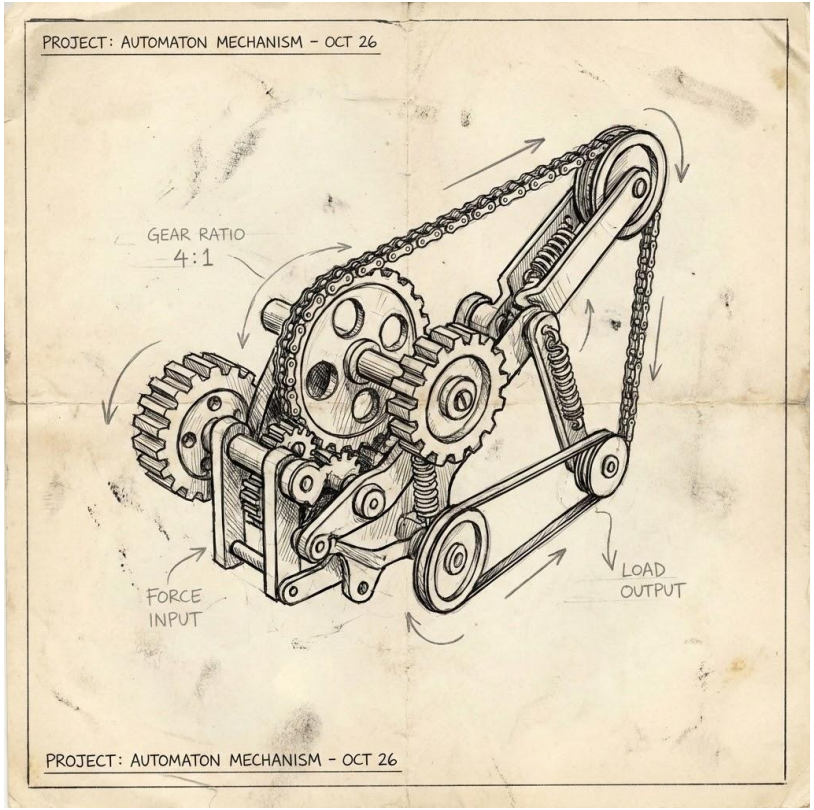

Mahindra 475 DI


WATTS & SONS
TEXTILES


ALLEGHENY STEEL & WIRE WORKS
EST. 1879
GENERAL ELECTRIC
POWER & LIGHT




Global supply chain
AI
Environmental sustainability metrics
60
Carbon footprint reduction
Energy efficiency
-50%
Energy efficiency
AGV



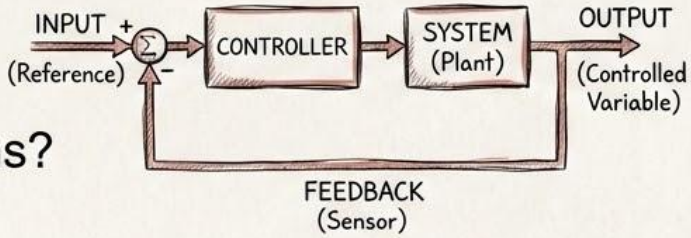
Control Systems

Control System

- A control system manages, commands, or regulates the behaviour of devices or systems using control loops to achieve a desired output.

- What are the components of a control system?

- What are some classifications of control systems?



Open and Closed Loop Control Systems





Open loop control systems

No feedback, fixed operation based on input





Closed loop control systems

Uses feedback to adjust output and maintain desired state